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In [1]: import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.cluster.hierarchy import linkage, dendrogram, set_link_color_palette
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.mixture import GaussianMixture
from sklearn.metrics import confusion_matrix
from scipy.optimize import linear_sum_assignment

# Standard vivid colors for the Dendrogram to separate it from the pastel dashbo
set_link_color_palette(['#1f77b4', '#ff7f0e', '#2ca02c', '#9467bd', '#8c564b', '

# =====
# 1. NORMALIZE
# =====
df = pd.read_csv("Data.csv")
df_normalized = df.copy()

numerical_cols = [
    'age', 'daily_screen_time_min', 'social_media_time_min',
    'daily_screen_time_without_social_media_min', 'social_media_ratio',
    'negative_interactions_count', 'positive_interactions_count',
    'sleep_hours', 'physical_activity_min', 'activity_to_screen_ratio',
    'anxiety_level', 'stress_level', 'mood_level'
]

scaler = MinMaxScaler()
for col in numerical_cols:
    df_normalized[col + '_norm'] = scaler.fit_transform(df_normalized[[col]])

# =====
# 2. ENCODE
# =====
df_encoded = df_normalized.copy()
mental_state_mapping = {'Healthy': 0, 'Stressed': 1, 'At_Risk': 2}
df_encoded['mental_state_encoded'] = df_encoded['mental_state'].map(mental_state
df_encoded = pd.get_dummies(df_encoded, columns=['platform', 'gender'], drop_fir

# =====
# 3. MANUAL DIMENSIONALITY REDUCTION (CUSTOM INDEXES + PLATFORMS)
# =====
df_encoded['Digital_Habits_Index'] = (df_encoded['daily_screen_time_min_norm'] +
df_encoded['Mental_Strain_Index'] = (df_encoded['stress_level_norm'] + df_encode

one_hot_platforms = [col for col in df_encoded.columns if col.startswith('platfo
manual_features = ['Digital_Habits_Index', 'Mental_Strain_Index'] + one_hot_plat

print("\n" + "="*50)
print("--- Manual Dimensionality Reduction Explanation ---")
print("Axis X (Digital_Habits_Index) = 50% Daily Screen Time + 50% Social Media
print("Axis Y (Mental_Strain_Index) = 50% Stress Level + 50% Anxiety Level")
print("Additional Features Included = All One-Hot Encoded Social Media Platform
print("="*50 + "\n")

```

file:///C:/Users/shaul/Desktop/קוד/התנסות/ב/פרויקט/ב/סמסטר/ב/לימודים/שנה/ב/Clustering.html

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plt.show()

# --- B. Execute Algorithms ---
df_clusters = pd.DataFrame(index=df_encoded.index)

kmeans = KMeans(n_clusters=num_clusters, random_state=42, n_init=10)
df_clusters['KMeans_Labels'] = kmeans.fit_predict(df_encoded[feature_list])

hierarchical = AgglomerativeClustering(n_clusters=num_clusters)
df_clusters['Hierarchical_Labels'] = hierarchical.fit_predict(df_encoded[fea

dbscan = DBSCAN(eps=0.5, min_samples=5)
df_clusters['DBSCAN_Labels'] = dbscan.fit_predict(df_encoded[feature_list])

gmm = GaussianMixture(n_components=num_clusters, random_state=42)
df_clusters['GMM_Labels'] = gmm.fit_predict(df_encoded[feature_list])

# Aligning GMM and KMeans Labels to match Hierarchical Labels dynamically
cm_gmm = confusion_matrix(df_clusters['Hierarchical_Labels'], df_clusters['G
row_ind_gmm, col_ind_gmm = linear_sum_assignment(-cm_gmm)
gmm_map = {col_ind_gmm[i]: row_ind_gmm[i] for i in range(len(row_ind_gmm))}
df_clusters['GMM_Labels'] = df_clusters['GMM_Labels'].map(gmm_map)

cm_km = confusion_matrix(df_clusters['Hierarchical_Labels'], df_clusters['KM
row_ind_km, col_ind_km = linear_sum_assignment(-cm_km)
km_map = {col_ind_km[i]: row_ind_km[i] for i in range(len(row_ind_km))}
df_clusters['KMeans_Labels'] = df_clusters['KMeans_Labels'].map(km_map)

# --- C. Scatter Plots ---
fig, axes = plt.subplots(2, 2, figsize=(16, 10))
fig.suptitle(f'Scatter Plots - {run_name}', fontsize=16, fontweight='bold')

sns.scatterplot(data=df_encoded, x=x_col, y=y_col, hue=df_clusters['KMeans_L
axes[0, 0].set_title('K-Means')
axes[0, 0].legend(bbox_to_anchor=(1.02, 1), loc='upper left', fontsize=8, ti

sns.scatterplot(data=df_encoded, x=x_col, y=y_col, hue=df_clusters['Hierarch
axes[0, 1].set_title('Hierarchical')
axes[0, 1].legend(bbox_to_anchor=(1.02, 1), loc='upper left', fontsize=8, ti

sns.scatterplot(data=df_encoded, x=x_col, y=y_col, hue=df_clusters['DBSCAN_L
axes[1, 0].set_title('DBSCAN')
axes[1, 0].legend(bbox_to_anchor=(1.02, 1), loc='upper left', fontsize=8, ti

sns.scatterplot(data=df_encoded, x=x_col, y=y_col, hue=df_clusters['GMM_Labe
axes[1, 1].set_title('GMM')
axes[1, 1].legend(bbox_to_anchor=(1.02, 1), loc='upper left', fontsize=8, ti

plt.subplots_adjust(left=0.05, right=0.88, wspace=0.45, hspace=0.3)
plt.show()

# --- D. Profiling Dashboards ---
df_profile = df.copy()

# Drop prefixes to save space if there are more than 3 clusters
if num_clusters > 3:
    hierarchical_names = [chr(65+i) for i in range(num_clusters)] # A, B, C,
    gmm_names = [str(i+1) for i in range(num_clusters)] # 1, 2, 3, 4...
else:
    hierarchical_names = [f'Group {chr(65+i)}' for i in range(num_clusters)]

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gmm_names = [f'Segment {i+1}' for i in range(num_clusters)] # Segment 1,

hier_map = {i: hierarchical_names[i] for i in range(num_clusters)}
gmm_map_names = {i: gmm_names[i] for i in range(num_clusters)}

df_profile['Hierarchical_Named'] = df_clusters['Hierarchical_Labels'].map(hier_map)
df_profile['GMM_Named'] = df_clusters['GMM_Labels'].map(gmm_map_names)

time_features = ['daily_screen_time_min', 'social_media_time_min', 'daily_screen_time_max', 'social_media_time_max']
psycho_features = ['anxiety_level', 'stress_level', 'mood_level']
interaction_features = ['negative_interactions_count', 'positive_interactions_count']
health_features = ['age', 'physical_activity_min', 'sleep_hours']

def plot_dashboard(cluster_col, algo_name, exact_order):
    fig, axes_dash = plt.subplots(2, 3, figsize=(25, 12))
    fig.suptitle(f'{run_name} | {algo_name} - Full Profiling Dashboard', fontweight='bold')

    def add_num_subplot(ax_obj, features, title, ylabel):
        df_melted = df_profile.melt(id_vars=[cluster_col], value_vars=features)
        df_melted['Metric'] = df_melted['Metric'].str.replace('_', ' ').str.title()
        sns.barplot(data=df_melted, x=cluster_col, y='Value', hue='Metric', ax_obj=ax_obj)
        ax_obj.set_title(title, fontsize=14, fontweight='bold')
        ax_obj.set_ylabel(ylabel)
        ax_obj.set_xlabel('')
        ax_obj.legend(bbox_to_anchor=(1.02, 1), loc='upper left', fontsize=7)

    if show_labels:
        for container in ax_obj.containers:
            ax_obj.bar_label(container, fmt='%.1f', padding=3, fontsize=7)

    add_num_subplot(axes_dash[0, 0], time_features, 'Time Metrics', 'Minutes')
    add_num_subplot(axes_dash[0, 1], psycho_features, 'Psychological Scores', 'Score')
    add_num_subplot(axes_dash[0, 2], interaction_features, 'Interactions Count', 'Count')
    add_num_subplot(axes_dash[1, 0], health_features, 'Age, Activity & Sleep', 'Hours')

    # Platform Countplot
    sns.countplot(data=df_profile, x=cluster_col, hue='platform', palette=da
    axes_dash[1, 1].set_title('Platform Preference', fontsize=14, fontweight='bold')
    axes_dash[1, 1].set_xlabel('')
    axes_dash[1, 1].legend(title='Platform', bbox_to_anchor=(1.02, 1), loc='upper left', fontsize=7)

    # Mental State Countplot
    sns.countplot(data=df_profile, x=cluster_col, hue='mental_state', palette=da
    axes_dash[1, 2].set_title('Mental State Distribution', fontsize=14, fontweight='bold')
    axes_dash[1, 2].set_xlabel('')
    axes_dash[1, 2].legend(title='Mental State', bbox_to_anchor=(1.02, 1), loc='upper left', fontsize=7)

    sns.despine()
    plt.subplots_adjust(left=0.04, right=0.88, wspace=0.55, hspace=0.35)
    plt.show()

plot_dashboard('Hierarchical_Named', 'Hierarchical', hierarchical_names)
plot_dashboard('GMM_Named', 'GMM', gmm_names)

# =====
# 6. RUNNING THE PIPELINES
# =====
sns.set_theme(style="whitegrid")

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```
# Run Manual Features with exactly 6 clusters and data Labels hidden
run_clustering_pipeline(run_name="Manual Dimensionality Reduction",
                        feature_list=manual_features,
                        x_col='Digital_Habits_Index',
                        y_col='Mental_Strain_Index',
                        scatter_palette='Set1',
                        dashboard_palette='Pastel1',
                        num_clusters=6,
                        show_labels=False)

# Leave PCA runs firmly at 3 clusters with data Labels shown
run_clustering_pipeline(run_name="PCA (2 Components)",
                        feature_list=pca2_features,
                        x_col='PCA2_1',
                        y_col='PCA2_2',
                        scatter_palette='Set1',
                        dashboard_palette='Pastel1',
                        num_clusters=3,
                        show_labels=True)

run_clustering_pipeline(run_name="PCA (3 Components)",
                        feature_list=pca3_features,
                        x_col='PCA3_1',
                        y_col='PCA3_2',
                        scatter_palette='Set1',
                        dashboard_palette='Pastel1',
                        num_clusters=3,
                        show_labels=True)
```

```
=====
--- Manual Dimensionality Reduction Explanation ---
Axis X (Digital_Habits_Index) = 50% Daily Screen Time + 50% Social Media Time
Axis Y (Mental_Strain_Index) = 50% Stress Level + 50% Anxiety Level
Additional Features Included = All One-Hot Encoded Social Media Platforms
=====
```

Total variance explained by 2 components: 0.48
 Total variance explained by 3 components: 0.61

```
=====
--- Top 5 features influencing PCA (2 Components) ---
```

PCA1:

stress_level_norm	0.328721
anxiety_level_norm	0.312361
mood_level_norm	0.297404
sleep_hours_norm	0.297372
physical_activity_min_norm	0.296037

Name: PCA1, dtype: float64

PCA2:

social_media_ratio_norm	0.480006
platform_WhatsApp	0.393346
daily_screen_time_without_social_media_min_norm	0.369641
platform_TikTok	0.294527
sleep_hours_norm	0.257793

Name: PCA2, dtype: float64

```
--- Top 5 features influencing PCA (3 Components) ---
```

PCA1:

stress_level_norm	0.328721
anxiety_level_norm	0.312361
mood_level_norm	0.297404
sleep_hours_norm	0.297372
physical_activity_min_norm	0.296037

Name: PCA1, dtype: float64

PCA2:

social_media_ratio_norm	0.480006
platform_WhatsApp	0.393346
daily_screen_time_without_social_media_min_norm	0.369641
platform_TikTok	0.294527
sleep_hours_norm	0.257793

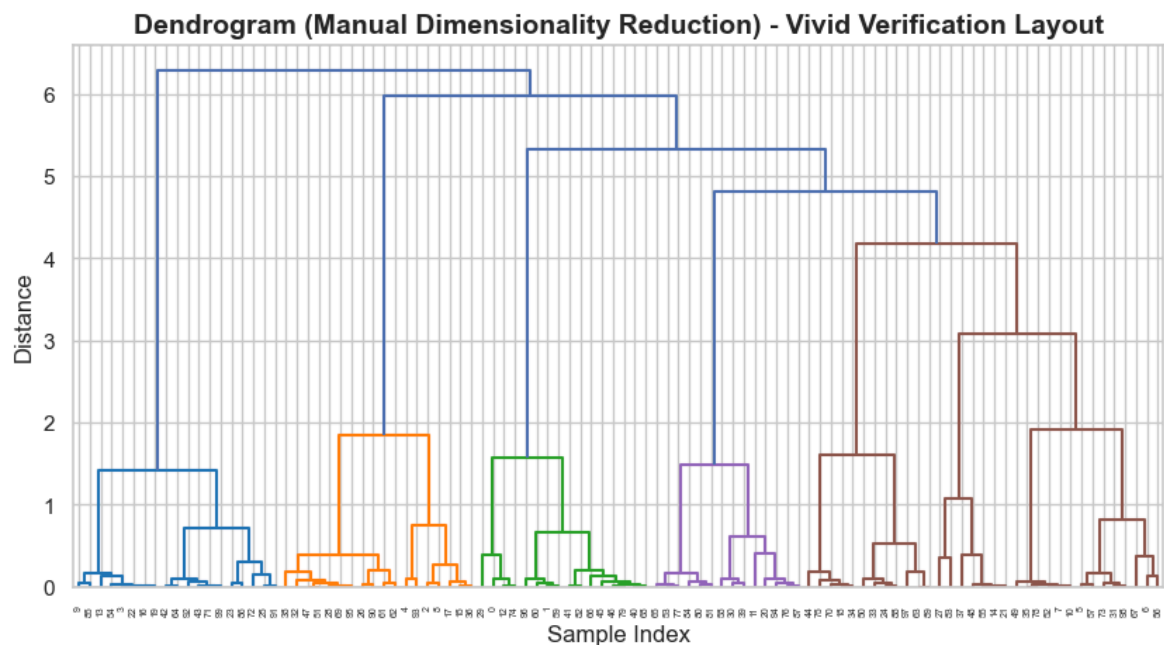
Name: PCA2, dtype: float64

White PCA3:

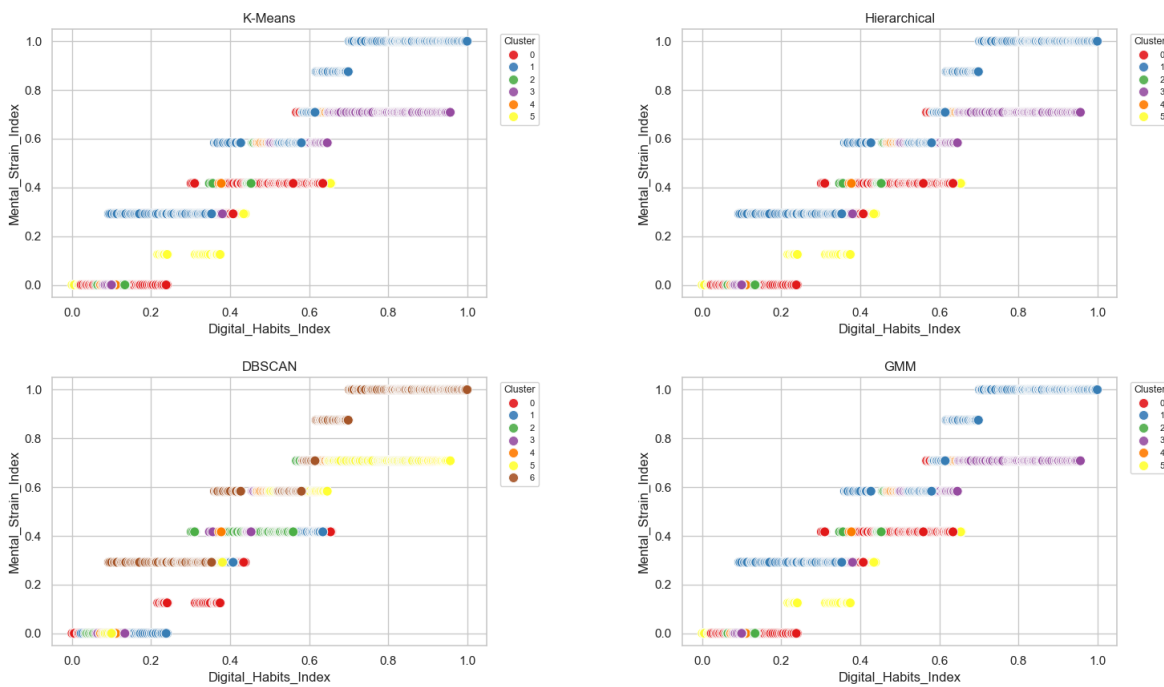
gender_Male	0.997754
gender_Other	0.041768
platform_Snapchat	0.033611
platform_WhatsApp	0.026058
social_media_ratio_norm	0.016710

Name: PCA3, dtype: float64

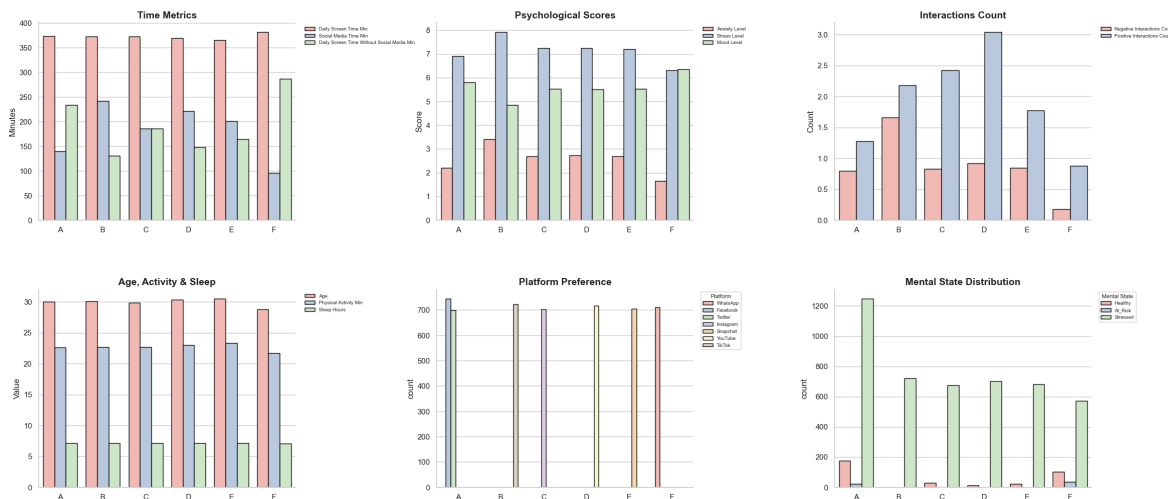
```
=====
EXECUTING PIPELINE: Manual Dimensionality Reduction (6 Clusters)
=====
```



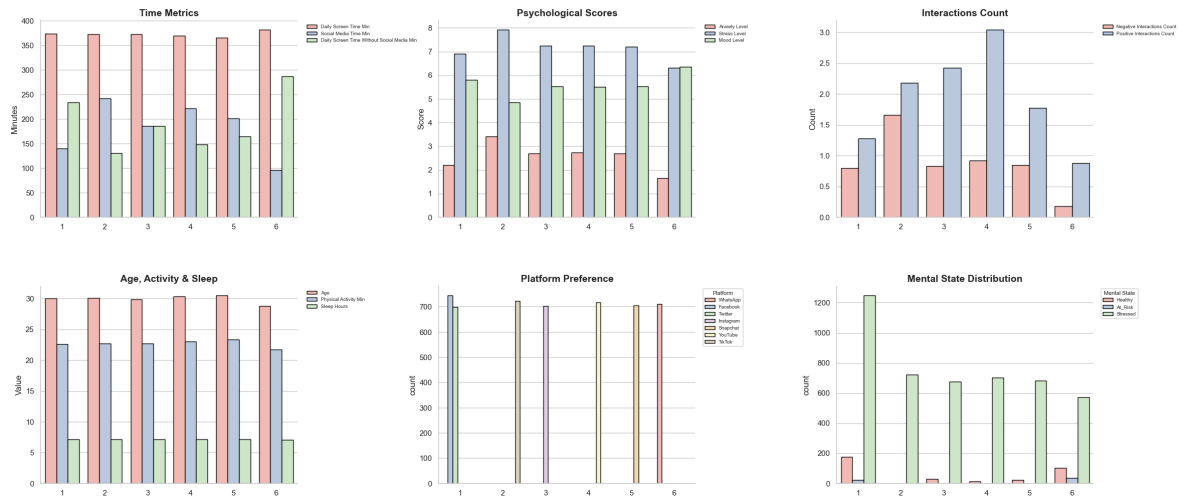
Scatter Plots - Manual Dimensionality Reduction



Manual Dimensionality Reduction | Hierarchical - Full Profiling Dashboard

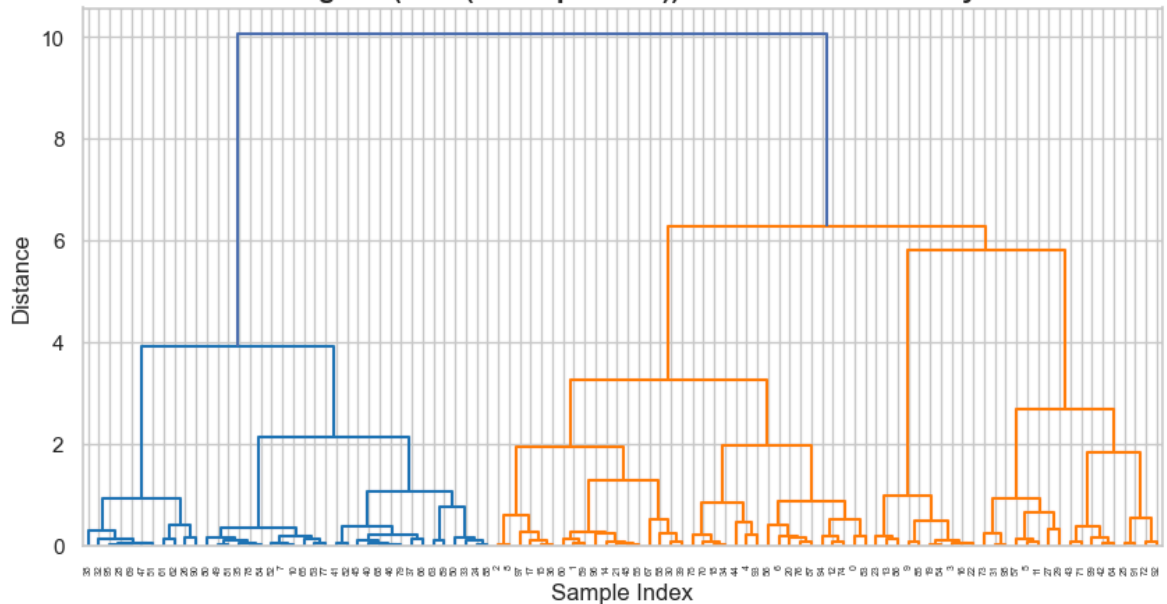


Manual Dimensionality Reduction | GMM - Full Profiling Dashboard

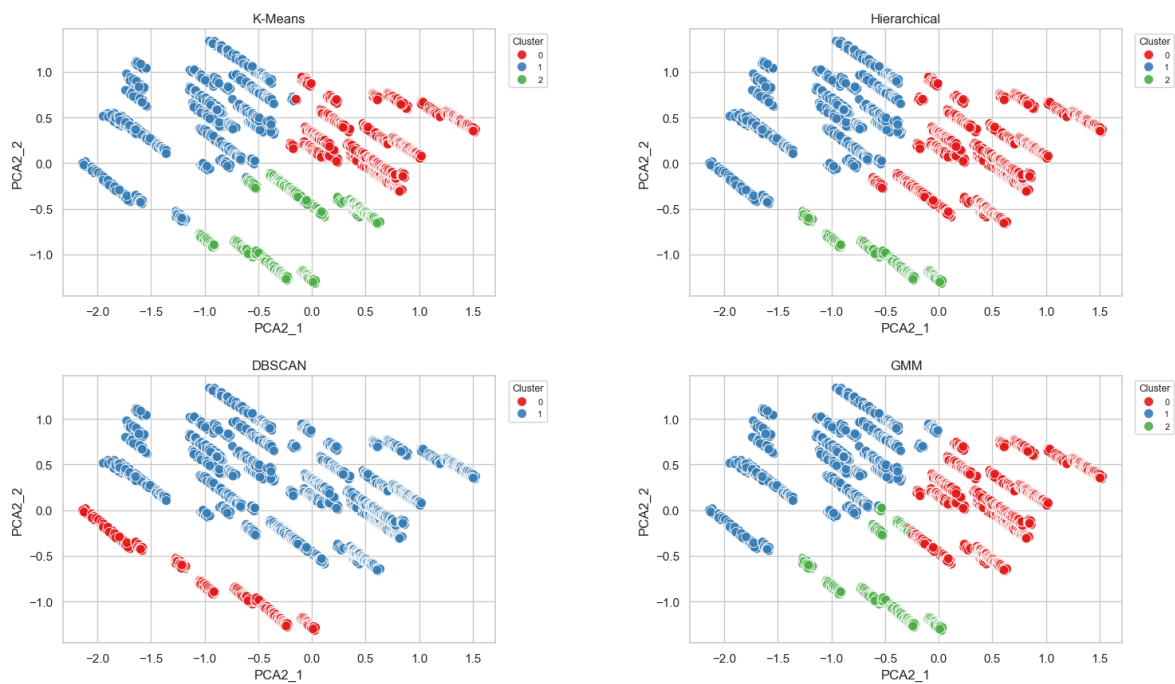


EXECUTING PIPELINE: PCA (2 Components) (3 Clusters)

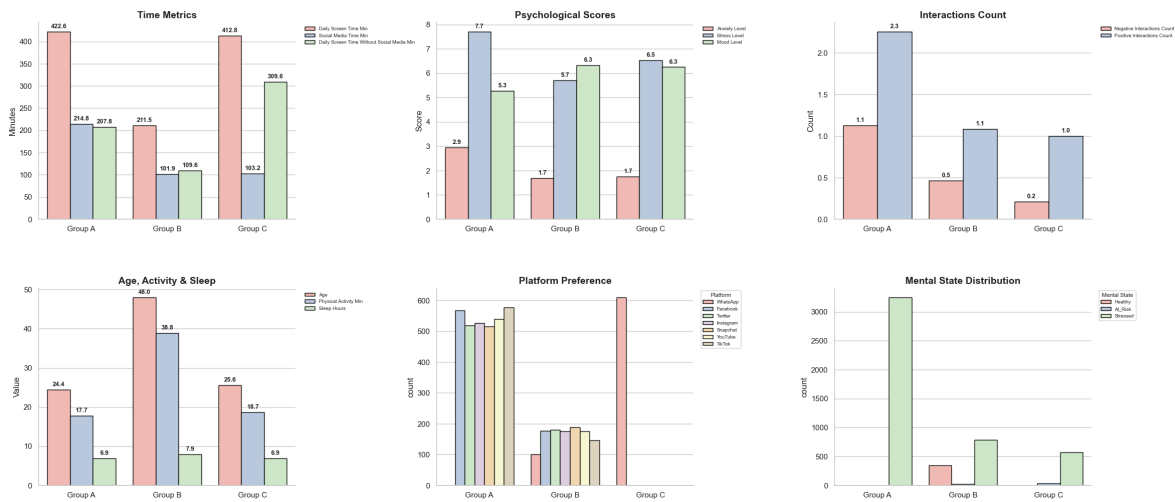
Dendrogram (PCA (2 Components)) - Vivid Verification Layout



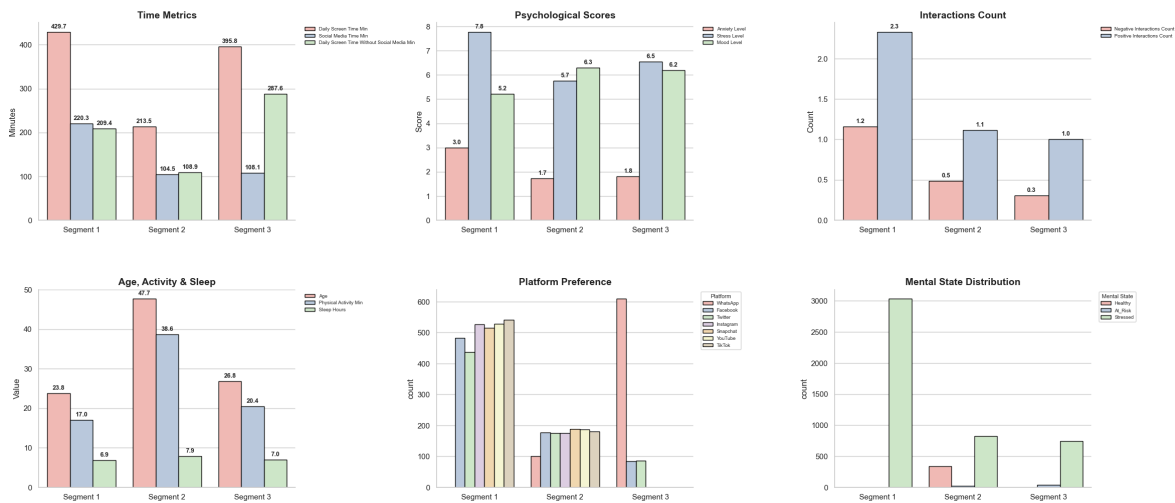
Scatter Plots - PCA (2 Components)



PCA (2 Components) | Hierarchical - Full Profiling Dashboard



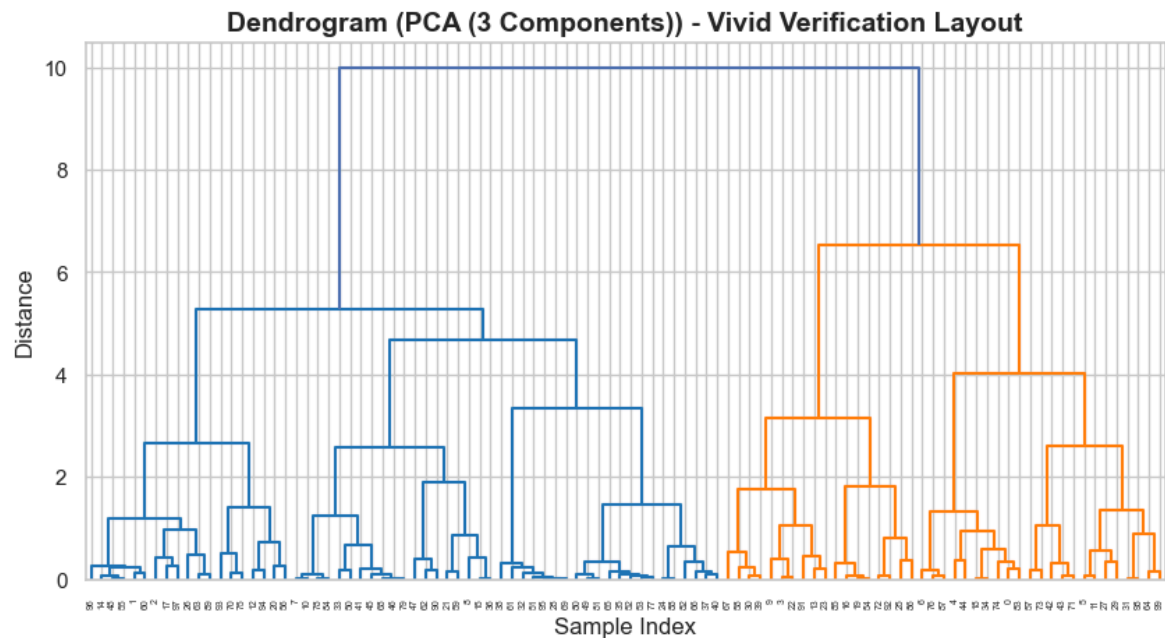
PCA (2 Components) | GMM - Full Profiling Dashboard



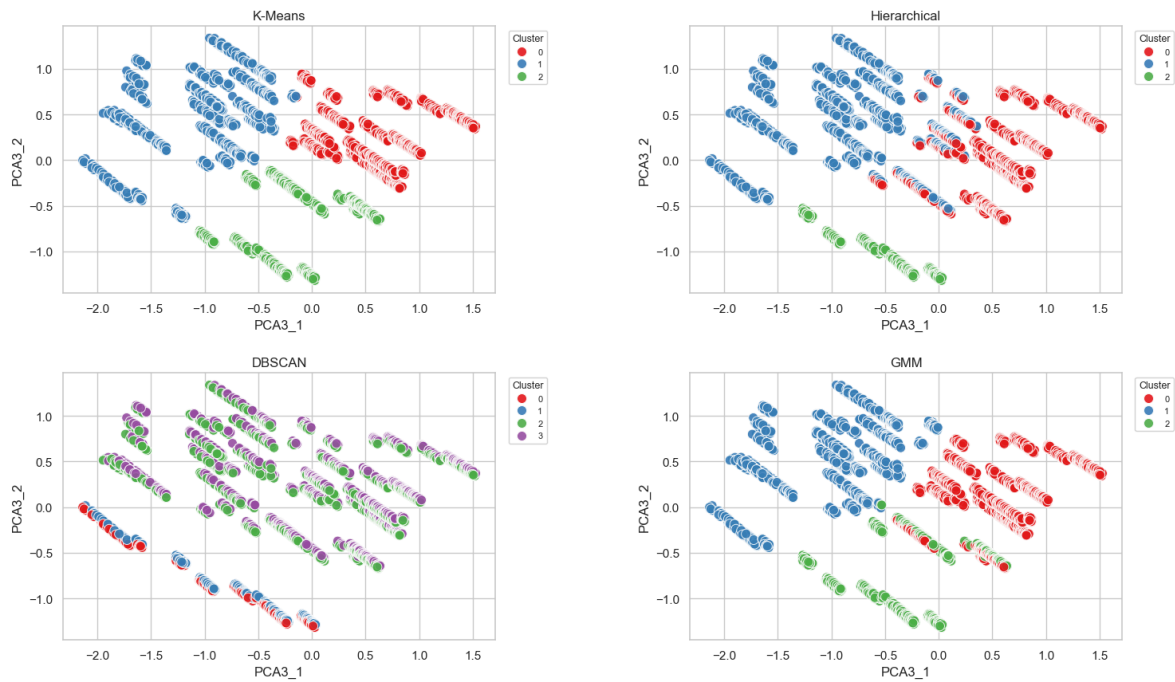
=====

EXECUTING PIPELINE: PCA (3 Components) (3 Clusters)

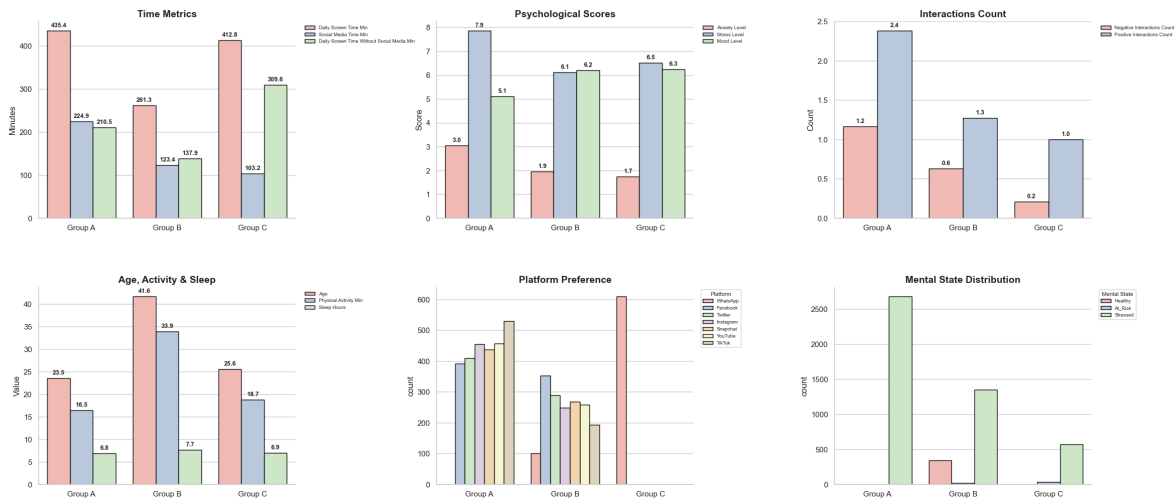
=====



Scatter Plots - PCA (3 Components)



PCA (3 Components) | Hierarchical - Full Profiling Dashboard



PCA (3 Components) | GMM - Full Profiling Dashboard

